

Student Performance Prediction Project

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Project Title: Predicting Student Academic Success Using Machine Learning

Course: ISOM 835 - Predictive Analytics and Machine Learning

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Introduction & Dataset Description

The goal of this project is to build predictive models that determine whether a secondary school student will pass or fail based on a variety of demographic, behavioral, and academic features. Predictive analytics in education can assist administrators and teachers in identifying at-risk students and designing early intervention strategies.

The dataset used in this project is the *Student Performance Data Set* from the UCI Machine Learning Repository. It includes detailed records of students in Portuguese secondary schools, with features related to family background, alcohol consumption, school support, and academic performance. The data has 33 features and approximately 650 instances.

Exploratory Data Analysis (EDA)

EDA was conducted using Python libraries such as pandas, matplotlib, and seaborn.

Key Insights:

- The target variable, G3, is the final grade. A binary column `pass_fail` was created to indicate pass (≥ 10) or fail (< 10).
- Features such as past failures, study time, and absences showed strong relationships with performance.
- Alcohol use (weekday and weekend) negatively correlated with grades.
- Heatmaps and scatter plots were used to visualize relationships between features.

Preprocessing

Steps Taken:

- Verified no missing data.

- Applied one-hot encoding to all categorical variables.
- Created a binary pass_fail column.
- Dropped G1, G2, and G3 to prevent data leakage.
- Performed an 80/20 train-test split with stratification.
- Standardized features using StandardScaler.

Business Questions

Q1: What student background and behavioral factors best predict whether a student will pass or fail?

Supports early intervention efforts by highlighting critical risk indicators.

Q2: Can we build a predictive model to classify students as likely to pass or fail?

Enables real-time flagging of at-risk students using accessible data.

Q3: What are the top 3 features that most impact student performance?

Helps educators focus on the most influential factors.

Predictive Modeling

Three models were trained: Logistic Regression, Random Forest, and XGBoost.

Model Comparison Table:

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	78.5%	80.1%	76.3%	78.1%
Random Forest	85.2%	86.0%	84.1%	85.0%
XGBoost	86.7%	88.1%	85.6%	86.8%

Feature Importance:

Top predictors of success:

- Number of past class failures
- Weekly study time
- Number of absences
- Parental education
- Alcohol consumption

Insights & Recommendations

- The XGBoost model showed the highest performance and would be ideal for implementation.
- Past failures, study time, and absences are the top predictors—schools can track these easily.
- Predictive systems could notify counselors or trigger interventions for students flagged as high-risk.
- Alcohol use and lack of parental involvement are soft indicators of disengagement.

Recommendations:

- Use predictive modeling to automate risk identification.
- Educate staff on key warning signs.
- Provide targeted tutoring and parental outreach for at-risk groups.

Ethics & Interpretability

While effective, these models raise ethical concerns:

- **Bias:** Variables such as parental education may reflect socioeconomic disparities.
- **Transparency:** Tree-based models are harder to interpret. Tools like SHAP should be used to explain results.
- **Accountability:** Human review must accompany predictive alerts to prevent unfair treatment.

Prediction systems must be explainable and audited regularly to avoid reinforcing inequality.

Appendix

Files & Folders:

- notebooks/ISOM835_EDA_Modeling.ipynb
- visualizations/pass_fail_distribution.png, correlation_heatmap.png
- report/Final_Report.pdf
- GitHub README and repository structure as specified

Colab Link:

 ISOM 835 Term Project-MARLON WESH

GitHub Repository:

<https://github.com/mtwesh/student-performance-prediction-ISOM835>